

VƯƠNG VIỆT HÙNG

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Hanoi, Vietnam - 11025, Vietnam

EDUCATION

- **Soongsil University** March 2024 - March 2026
Master's Degree in Electronic Engineering Seoul, South Korea
 - GPA: 4.24/4.50
- **Hanoi University of Science and Technology** August 2019 - September 2023
Bachelor's Degree in Control Engineering and Automation Hanoi, Vietnam
 - GPA: 3.34/4.00

EXPERIENCE

- **VinDynamics** March 2026 – current
SLAM & Navigation Engineer Hanoi, Vietnam

Developed and deployed SLAM and height mapping systems for humanoid robotic platforms, with a focus on sensor integration, system evaluation, and optimization.
- **ANDA Lab, Soongsil University** March 2024 – March 2026
Research Assistant Seoul, South Korea

Conducted research in LiDAR panoptic segmentation, scene completion, domain adaptation, and continual learning; contributed to peer-reviewed publications, researcher mentoring, and technical seminars.
- **Rikai Mind** August 2022 – January 2023
AI Engineer Intern Hanoi, Vietnam

Developed and deployed a computer vision system for object volume estimation from monocular images and camera-object distance.
- **Mandevices Laboratory, Hanoi University of Science and Technology** August 2020 – September 2023
Research Assistant Hanoi, Vietnam

Conducted research in deep-learning-based bearing fault diagnosis and contributed to data collection, publications, and researcher mentoring.

PUBLICATIONS

- **Domain Adaptive LiDAR Panoptic Segmentation via Pseudo-Supervised Instance Learning** May 2026
First author – IEEE Transactions on Intelligent Transportation Systems, Early Access
DOI: [10.1109/TITS.2026.3695516](https://doi.org/10.1109/TITS.2026.3695516)
 - Proposed PILLA, a domain adaptation framework that generates instance-level pseudo-labels via unsupervised clustering and applies a noise-aware loss for robust pseudo-supervision.
 - Achieved 35.97% PQ on nuScenes in cross-dataset adaptation from SemanticKITTI, outperforming AdaptLPS by 13.14 percentage points.
- **Instance Embedding as Queries for DETR-based LiDAR Panoptic Segmentation** October 2025
Co-first author – IEEE Transactions on Intelligent Vehicles, Vol. 10, Issue 10, pp. 4629 - 4641
DOI: [10.1109/TIV.2024.3488035](https://doi.org/10.1109/TIV.2024.3488035)
 - Proposed a DETR-based architecture that uses instance embeddings as object queries for LiDAR panoptic segmentation.
 - Achieved state-of-the-art performance with 63.40% PQ on SemanticKITTI benchmark and 77.08% PQ on nuScenes benchmark, outperforming previous methods by 1.20 and 1.94 percentage points, respectively.
- **FuPaSCo: Long-range and Local Context Fusion for 3D Panoptic Scene Completion** November 2025
Co-author – Image and Vision Computing, Vol. 163, pp. 105776
DOI: [10.1016/j.imavis.2025.105776](https://doi.org/10.1016/j.imavis.2025.105776)

- Developed FuPaSCo, a dual-branch CNN–Transformer framework that combines local and long-range features through feature fusion and alignment for LiDAR panoptic scene completion.
- Achieved 26.10% $PQ^\dagger$ on SemanticKITTI benchmark, outperforming the previous best method PasCo by 2.49 percentage points.

• **Generalized Simulation-Based Domain Adaptation Approach for Intelligent Bearing Fault Diagnosis**

July 2024

Co-author – *Arabian Journal for Science and Engineering*, Vol. 49, Issue 12, pp. 16941–16957
DOI: [10.1007/s13369-024-09282-1](https://doi.org/10.1007/s13369-024-09282-1)

- Developed a simulation-based domain adaptation approach that generates synthetic vibration signals from bearing geometry to address data scarcity in intelligent bearing fault diagnosis.
- Achieved 74–98% fault classification accuracy across different bearing types using a CNN trained on synthesized vibration signals.

• **ConvMamba: A Data-Efficient Neural Network for Bearing Fault Diagnosis**

July 2024

Co-author – 7th International Conference on Green Technology and Sustainable Development (GTSD), pp. 165–172. IEEE. 2024-07-25, Ho Chi Minh City, Vietnam
DOI: [10.1109/GTSD62346.2024.10674880](https://doi.org/10.1109/GTSD62346.2024.10674880)

- Developed ConvMamba, a data-efficient architecture combining convolutional networks and Vision Mamba for bearing fault diagnosis.
- Improved classification accuracy by 27 percentage points over the predecessor Hybrid-CNN-MLP model on the CWRU bearing fault dataset.

MANUSCRIPTS UNDER REVIEW

• **SM3-PSC: Sparse Multi-Modal Mamba for 3D Panoptic Scene Completion with Multi-Scale Feature Refinement**

2026

Co-author – Manuscript submitted to *IEEE Transactions on Image Processing*

- Developed SM3-PSC, a sparse multi-modal framework that fuses LiDAR and camera features and employs Mamba-based multi-scale feature refinement for 3D panoptic scene completion.
- Achieved 29.20% $PQ^\dagger$ on SemanticKITTI, outperforming the previous best method by 3.10 percentage points through improved completion of both object instances and surrounding regions.

• **LW-PSC: Lightweight Design for LiDAR Panoptic Scene Completion**

2026

Co-author – Manuscript submitted to *Applied Soft Computing*

- Developed LW-PSC, a lightweight BEV-based framework that combines Cartesian and Polar representations with center-focused feature refinement for real-time LiDAR panoptic scene completion.
- Achieved 13.59% PQ at 11.3 FPS on SemanticKITTI, outperforming PaSCo by 1.47 percentage points in PQ while reducing the model size from 40.65M to 29.16M parameters.

PROJECTS

• **Visual SLAM Profiling and Latency Optimization**

August 2026

VinDynamics

- Profiled the OKVIS2 visual SLAM pipeline, identifying landmark matching and local-window optimization as major bottlenecks, and evaluated low-latency front-end pose publishing with only 1.16 cm translational APE RMSE relative to back-end poses.
- Redesigned frame scheduling to prioritize the latest available frame and drop stale frames, reducing average front-end latency from 124.00 ms to 47.00 ms and back-end latency from 164.30 ms to 87.64 ms.

• **Multi-Sensor Local Height Mapping for Humanoid Robot**

July 2026 – August 2026

VinDynamics

- Extended a LiDAR-based GPU height mapper to integrate Livox LiDAR and Intel RealSense depth data, implementing sensor synchronization, rate limiting, pose lookup, and robot self-filtering for deployment on an in-house humanoid robot.

- Implemented and benchmarked pose extrapolation to compensate for SLAM latency, reducing translational APE RMSE from 2.148 cm to 0.445 cm (79.3%) for a 30 Hz depth camera using linear extrapolation.

- **Scene Flow-Guided Online Continual LiDAR Panoptic Tracking**

September 2025 – February 2026

Graduation thesis, Soongsil University

- Proposed SF-LPT, the first online continual learning framework for LiDAR panoptic tracking, using self-supervised scene flow for continuous adaptation and temporal instance association across sequential scans.
- Achieved 16.45% LSTQ on SemanticKITTI in cross-dataset continual adaptation from nuScenes, outperforming CoDEPS by 5.87 percentage points.

- **Stereo Matching and 3D Reconstruction**

March 2023 – August 2023

Graduation thesis, Hanoi University of Science and Technology

- Implemented and compared block matching and deep learning approaches for dense stereo matching on SceneFlow (EPE: 77.32 px, D1: 79.04%).
- Reconstructed 3D scenes from estimated disparity maps via triangulation and deployed the pipeline on a Jetson Nano-based wheeled robot.

- **Object Volume Estimation from Images**

August 2022 – December 2022

Rikai Mind

- Developed an object volume estimation model from monocular images and camera–object distance, achieving an R^2 score of 84.78%.
- Extracted visual embeddings with CNNs, localized objects using keypoint regression, and fused visual features with camera–object distance for volume regression; deployed the model using FastAPI.

SKILLS

- **Programming languages:** Python, C/C++, MATLAB
- **Frameworks:** PyTorch, TensorFlow, OpenCV, ROS2, Kalibr (for sensor calibration)
- **Hardware platforms:** Jetson Orin Nano, Jetson Nano, Raspberry Pi
- **Sensors:** Intel RealSense camera, ZED X camera, Livox LiDAR
- **Development and deployment tools:** Git, GitHub, GitLab, Docker, FastAPI, Milvus
- **Others:** LaTeX

LANGUAGES

- **English:** IELTS 7.5
- **Vietnamese:** Native speaker